

Numerical Solution of a Slow-Fast PDE Model for Nitrifying Bacteria Biofilms: Methods, Validation, Extensions, and Results

Contents

1	Introduction and Scope	3
2	The Biological System and the Governing Equations	3
2.1	The biology	3
2.2	The state variables and equations	4
2.3	Why substrate and bacterial equations cannot be solved together naively . .	5
3	The Central Numerical Strategy: Separating Fast and Slow Dynamics	5
3.1	The scale-separation parameter	5
3.2	The reduction	5
3.3	Non-dimensionalisation	6
4	Parameters	7
4.1	Oxidation reactions and stoichiometry	7
4.2	Parameter tables	7
5	Numerical Methods: Design and Reasoning	9
5.1	Spatial discretisation: finite volume method	9
5.2	Handling the radial coordinate singularity at $r = 0$	9
5.3	Solving the nonlinear substrate system: Newton's method	10
5.4	A second solver explored then abandoned: Picard iteration	10
5.5	A third solver: pseudo-transient continuation	10
5.6	Time-stepping the bacterial equations: fully implicit backward Euler	11
5.7	Advection discretisation: first-order upwinding	11

6	The Full Chronology: Problems Found, Diagnosed, and Fixed	11
6.1	The density-dependent death term	11
6.2	The reaction term incorrectly zeroed at the domain centre	11
6.3	The approximate cell-volume near the domain centre	12
6.4	Confirming second-order accuracy despite a floating-point illusion	12
6.5	Two errors in reaction-coefficient assembly	12
6.6	A boundary-condition bug affecting only nonzero flux	13
6.7	A physically implausible solution accepted by Newton	13
6.8	A self-inflicted sign error, and genuine degeneracy handled gracefully	13
6.9	A diagnostic-transparency problem from the fallback	14
6.10	A duplicated-computation bug in the bacterial solver	14
6.11	A remaining instance of the duplication pattern, now consolidated	14
6.12	The closed-form benchmark used throughout	15
7	Validation and Cross-Checking	15
8	The Two-Dimensional Extension	16
8.1	Motivation and design decision	16
8.2	Discretisation and validation	16
8.3	Sector formation reproduction	17
8.4	Investigating the sector-formation transition	17
9	Closing the Multi-Substrate Neumann Gap	18
9.1	The architectural problem	18
9.2	The refactor	18
9.3	Two further bugs found while building the fix	19
9.4	Result and honest remaining limitation	19
10	A Direct Quantitative Comparison Against the Source Paper	20
10.1	Reproducing the paper’s travelling-wave result	20
10.2	Re-verification with completely independent implementation	20
11	Present Status and Open Items	21
11.1	What is firmly established	21
11.2	What remains open	21
12	File-by-File Guide to the Codebase	22

1 Introduction and Scope

This document presents a complete account of the numerical work carried out on a mathematical model of nitrifying bacteria growing in a biofilm granule. It is designed to be self-contained and to answer, on its own, every question a reader would have about what was built, why each design choice was made, what went wrong during development and how it was fixed, what was verified and through what means, and what remains open.

The underlying biological and mathematical model is taken from Freingruber, Gonzalez-Cabaleiro and Yoldas (arXiv:2512.13156). That paper introduces a general cross-diffusion PDE framework for competing and cooperating microbial species. This work takes their nitrification case as a starting point, builds a complete numerical solver for it from scratch, subjects that solver to an extensive validation campaign, extends it to two spatial dimensions, closes a boundary-condition limitation left open partway through the project, and reproduces one of the source paper’s own quantitative numerical results independently.

The presentation moves from the biology and equations, through the numerical design choices and their physical justifications, through the full chronology of problems found and fixed, through validation, through the two-dimensional extension, through the boundary-condition refactor that closed out an open limitation, to the quantitative benchmark against the source paper, and finally to a complete file-by-file guide to the codebase. Throughout, the emphasis is on clarity: why each choice was made, what evidence supports each claim, and where genuine uncertainties remain.

2 The Biological System and the Governing Equations

2.1 The biology

Nitrification is the biological oxidation of ammonium to nitrate, carried out by three functional groups of bacteria.

Ammonia-oxidising bacteria (AOB) convert ammonium (NH_4^+) to nitrite (NO_2^-). Nitrite-oxidising bacteria (NOB) convert that nitrite to nitrate (NO_3^-). Comammox bacteria (CMX), discovered recently, can perform the entire conversion from ammonium to nitrate directly in a single step, bypassing the need for NOB.

These three species interact in two distinct ways. AOB and NOB form a commensal relationship: NOB depends entirely on the nitrite AOB produces as a metabolic by-product. Wherever AOB thrives, NOB finds its substrate nearby. AOB and CMX, by contrast, compete directly: both consume ammonium, so substrate consumed by one is unavailable to the

other. All three additionally require oxygen, so all three compete for oxygen regardless of their nitrogen source.

This structure – commensalism between one pair, competition between another, and universal competition for oxygen – is what makes this system spatially interesting. In a well-mixed reactor, none of the spatial organization described below would arise; species would simply average out based on bulk chemistry. In a biofilm granule, oxygen and ammonium enter from the surface and are consumed as they diffuse inward, creating a chemical environment that varies with depth. Near the surface, both oxygen and ammonium are plentiful; deep inside, oxygen is consumed faster than it can diffuse in, creating an anoxic interior. This spatial variation is the central phenomenon the model captures: the biofilm self-organizes into an oxygen-rich outer shell where all species can grow, and an oxygen-starved interior where growth ceases regardless of ammonium availability.

2.2 The state variables and equations

Seven quantities are tracked as functions of position and time: three bacterial population densities, u_{AOB} , u_{NOB} , u_{CMX} ; and four substrate concentrations, c_{NH_4} , c_{NO_2} , c_{NO_3} , c_{O_2} .

For each bacterial species i :

$$\frac{\partial u_i}{\partial t} = \nabla \cdot [d_i \nabla u_i + a_i u_i \nabla \rho] + u_i \left(r_i \prod_j \frac{c_j}{K_{ij} + c_j} - b_i \rho \right), \quad \rho = \sum_i u_i. \quad (1)$$

The first diffusion term, $d_i \nabla u_i$, represents ordinary bacterial spreading. The second term, $a_i u_i \nabla \rho$, represents bacteria being physically pushed out of crowded regions toward less crowded areas – a crowding-driven movement term that depends on the *total* biomass ρ , not just species i 's own density. This cross-diffusion coupling is what makes this system genuinely multispecies and nonlinear at the spatial level.

The growth and death terms, $r_i \prod_j c_j / (K_{ij} + c_j) - b_i \rho$, follow Monod kinetics. Each species needs two substrates simultaneously; if either is absent, growth is zero. The death term is density-dependent, giving the population a carrying capacity – a constant death rate would allow unbounded growth in rich regions.

For each substrate j :

$$\frac{\partial c_j}{\partial t} = D_j \Delta c_j + (\text{production by bacteria}) - (\text{consumption by bacteria}). \quad (2)$$

Substrates diffuse, are supplied at the boundary for ammonia and oxygen, and are produced or consumed by bacterial metabolism. Nitrite and nitrate exist only as metabolic

by-products; they have no external supply.

2.3 Why substrate and bacterial equations cannot be solved together naively

Substrate diffusivity ($\sim 10^{-4}$ m²/day) is four to five orders of magnitude larger than bacterial spreading rates. A single, uniform time step small enough to resolve oxygen diffusion ($\sim 10^{-5}$ seconds) would require $\sim 10^{10}$ steps to reach a single biologically meaningful day – computationally infeasible. Rather than fight this disparity, the entire numerical strategy exploits it.

3 The Central Numerical Strategy: Separating Fast and Slow Dynamics

3.1 The scale-separation parameter

Define $\varepsilon_j = r_{\max} L^2 / D_j$, the ratio of two timescales: how long substrate j takes to diffuse across the domain divided by how long the fastest-growing bacterial species takes to double. If ε_j is small, substrate j equilibrates long before bacteria have changed appreciably.

3.2 The reduction

Rescaling the substrate equation’s time derivative by ε_j and taking $\varepsilon_j \rightarrow 0$ removes the time derivative entirely, changing the equation’s type from parabolic to elliptic:

$$0 = D_j \Delta c_j + (\text{production}) - (\text{consumption}). \quad (3)$$

The substrate field at any moment can then be obtained directly by solving a purely spatial equation, given the current bacterial configuration:

$$u^n \longrightarrow \text{solve elliptic substrate system for } c^n \longrightarrow \text{advance } u^n \rightarrow u^{n+1} \longrightarrow \text{repeat}.$$

This reduction is valid only when ε_j truly is small. For every parameter set used in this project, ε_j was computed explicitly and came out in the range 10^{-3} to 10^{-2} – substrate diffusion is 200 to 1000 times faster than bacterial growth. A specific exception, where the source paper’s own published parameters place ε far outside this regime, is discussed in the two-dimensional extension below.

3.3 Non-dimensionalisation

Every dimensional quantity was rescaled into dimensionless form before any solver was built, for three reasons: raw dimensional values span enormous ranges, which is poor practice for Newton solver conditioning; non-dimensionalisation makes ε an explicit, computable number rather than an implicit assumption; and three parameter presets, differing hugely in magnitude, needed a common footing.

Reference scales. Length $L = 500$ micrometers, the biofilm's physical radius. Reference concentration c_{ref} , the ammonium feed concentration – ammonium is upstream of the whole system, every species consumes it directly or depends on it indirectly. Reference bacterial density u_{ref} , the physically packed biofilm density (500 kg/m^3 , $\approx 2.03 \times 10^7 \mu\text{M}$). An alternative, deriving u_{ref} from substrate supply and bacterial yield, was tried and rejected: it made the reaction term negligible compared to diffusion, preventing the oxygen-starved interior zone from forming at all. The packed-density choice instead gives reaction and diffusion terms of comparable size, consistent with genuine reaction-diffusion balance – a modelling choice confirmed to be load-bearing, not incidental. Reference times: slow time $\tau_{\text{slow}} = 1/r_{\text{max}}$; fast time $\tau_{\text{fast}} = L^2/D_{\text{NH}_4}$.

With lengths scaled by L , time by τ_{slow} , bacterial densities by u_{ref} , and concentrations by c_{ref} , the equations become:

$$\frac{\partial \hat{u}_i}{\partial \hat{t}} = \hat{D}_i \nabla^2 \hat{u}_i + \hat{A}_i \nabla \cdot (\hat{u}_i \nabla \hat{\rho}) + \hat{u}_i \left[\hat{r}_i M(\hat{c}_p; \hat{K}_{ip}) M(\hat{c}_{O_2}; \hat{K}_{iO_2}) - \hat{b}_i \hat{\rho} \right], \quad (4)$$

$$\varepsilon_j \frac{\partial \hat{c}_j}{\partial \hat{t}} = \nabla^2 \hat{c}_j + \varepsilon_j \hat{c}_j \hat{g}_j(\hat{u}, \hat{c}), \quad (5)$$

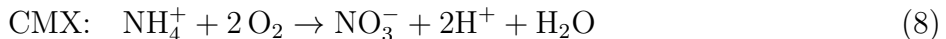
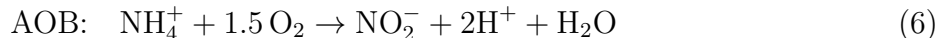
where $M(c; K) := c/(K + c)$, and dividing by ε_j and taking $\varepsilon_j \rightarrow 0$ recovers the elliptic reduction.

Key dimensionless groups: $\hat{r}_i, \hat{b}_i$ (growth and death relative to the fastest species), $\hat{K}_{ij}$ (half-saturation relative to reference concentration), $\hat{D}_i, \hat{A}_i$ (bacterial diffusion and crowding-movement coefficients), $Da_{ij} = u_{\text{ref}}/(Y_{ij}c_{\text{ref}})$ (Damkohler number, the ratio of reaction to diffusion capacity), and $\Lambda_{ij}, \Lambda_i^{\text{prod}}$ – the actual coefficients multiplying uptake and production terms, with $\Lambda_{ij} = \hat{r}_i \cdot \varepsilon_j \cdot Da_{ij}$. The distinction between Λ_{ij} (built with the diffusivity of the consumed substrate) and Λ_i^{prod} (built with the diffusivity of the produced substrate) turned out to matter significantly, as discussed below.

4 Parameters

Three parameter presets are used: toy (clean round numbers), rebecca (rough estimate from team correspondence), and eloi (literature-derived realistic parameters). All three pass every structural and correctness test, but this turned out to matter for how bugs were found: two consequential bugs were completely invisible on toy and rebecca, because both have near-uniform growth rates across species and, for toy, a single shared diffusivity across all substrates – exactly the conditions under which those errors cancel out. Only eloi, with realistic heterogeneity, exposed them. The general lesson: a clean, uniform preset is good for isolating one mechanism at a time, but is not sufficient on its own to catch bugs that only manifest under realistic heterogeneity.

4.1 Oxidation reactions and stoichiometry



These ratios were used to derive oxygen yield values for eloi, since the literature source does not report them directly.

4.2 Parameter tables

Parameter	Symbol	Units	toy	rebecca	eloi
AOB max growth rate	r_{AOB}	1/day	1.0	0.5	0.9489
NOB max growth rate	r_{NOB}	1/day	1.0	1.5	0.5030
CMX max growth rate	r_{CMX}	1/day	1.0	0.375	0.0692
AOB decay rate	b_{AOB}	1/day	0.1	0.05	0.0949
NOB decay rate	b_{NOB}	1/day	0.1	0.15	0.0503
CMX decay rate	b_{CMX}	1/day	0.1	0.0375	0.0069
AOB K_{NH_4}	K_{AOB, NH_4}	μM	1.0	1.0	1.453
AOB K_{O_2}	K_{AOB, O_2}	μM	1.0	3.13	3.606
NOB K_{NO_2}	K_{NOB, NO_2}	μM	1.0	1.0	0.00008894
NOB K_{O_2}	K_{NOB, O_2}	μM	1.0	3.13	2.594

Parameter	Symbol	Units	toy	rebeca	eloi
CMX K_{NH_4}	K_{CMX,NH_4}	μM	1.0	1.0	0.063
CMX K_{O_2}	K_{CMX,O_2}	μM	1.0	3.13	3.13
AOB yield (NH_4)	Y_{AOB,NH_4}	mol/mol	0.1	0.0409	0.0409
AOB yield (O_2)	Y_{AOB,O_2}	mol/mol	0.1	0.0614	0.0273
NOB yield (NO_2)	Y_{NOB,NO_2}	mol/mol	0.1	0.0242	0.0242
NOB yield (O_2)	Y_{NOB,O_2}	mol/mol	0.1	0.0121	0.0484
CMX yield (NH_4)	Y_{CMX,NH_4}	mol/mol	0.1	0.0651	0.0651
CMX yield (O_2)	Y_{CMX,O_2}	mol/mol	0.1	0.1302	0.0326
Bacterial diffusivity	d_i	m^2/day	1e-6	1e-6	1e-6

Table 1: Dimensional species parameters. The rebeca yield values shown here are already the shipped rescale (divided by 100) of the raw source values; see the sensitivity analysis below.

Parameter	Symbol	Units	toy	rebeca	eloi
NH_4 diffusivity	D_{NH_4}	m^2/day	1e-4	8.64e-5	1.692e-4
NO_2 diffusivity	D_{NO_2}	m^2/day	1e-4	8.64e-5	1.6512e-4
NO_3 diffusivity	D_{NO_3}	m^2/day	1e-4	8.64e-5	1.644e-4
O_2 diffusivity	D_{O_2}	m^2/day	1e-4	8.64e-5	1.8144e-4
NH_4 feed concentration	c_{∞,NH_4}	μM	500.0	554.0	500.0
NO_2 feed concentration	c_{∞,NO_2}	μM	0.0	0.0	0.0
NO_3 feed concentration	c_{∞,NO_3}	μM	0.0	0.0	0.0
O_2 feed concentration	c_{∞,O_2}	μM	200.0	187.5	93.8

Table 2: Dimensional substrate parameters.

Parameter	Symbol	Value
Domain radius	L	500 μm
Cross-diffusion to diffusion ratio	a_i/d_i	10.0
Packed biofilm density	–	500 kg/m^3
Biomass molecular weight	–	24.6 g/mol

Table 3: Domain and derived constants.

The cross-diffusion ratio $a_i/d_i = 10$ merits note. The source paper states $d_i \ll a_i$ qualitatively but gives no specific ratio in its general discussion. This value was chosen independently using order-of-magnitude reasoning; cross-referencing the paper’s own Table 1 later

found $d_i = 10^{-6}$ and $a_i = 10^{-5}$ there, a ratio of exactly 10, matching this project’s independently chosen value exactly. Because this is a direct match against the paper’s own stated numbers rather than a coincidence, this ratio is treated as confirmed against source.

Preset	ε_{NH_4}	ε_{NO_2}	ε_{NO_3}	ε_{O_2}	$\varepsilon = \max_j \varepsilon_j$
toy	–	–	–	–	2.5e-3
rebeca	–	–	–	–	4.3e-3
eloi	–	–	–	–	1.4e-3

Table 4: Scale-separation parameter, all presets. Blank cells to be filled from the codebase’s own `nondim.py` output.

5 Numerical Methods: Design and Reasoning

5.1 Spatial discretisation: finite volume method

The domain is divided into small control volumes. Rather than approximating derivatives at grid points directly, the discretisation integrates the flux balance exactly over each control volume:

$$\text{Lap}_i = \frac{\text{flux out, east face} - \text{flux out, west face}}{\text{volume of cell } i}. \quad (9)$$

This guarantees exact mass conservation at machine precision, at any resolution – an algebraic identity, not an asymptotic property, since summing this equation weighted by cell volume causes every internal flux to cancel identically, leaving only the outer boundary’s contribution. This was exploited throughout development as an unambiguous diagnostic; two real bugs were caught this way.

5.2 Handling the radial coordinate singularity at $r = 0$

In radial coordinates, the Laplacian is $(1/r^p)d/dr(r^p dc/dr)$, singular at $r = 0$ – a coordinate artifact, not a physical one. The standard approach integrates the flux balance directly over the control volume centered at $r = 0$, whose inner face has zero area, so the singularity never actually appears in the computation. Two separate mistakes were made around this singularity during development, described below.

5.3 Solving the nonlinear substrate system: Newton’s method

At each slow step the elliptic substrate system must be solved – a large coupled nonlinear system. Newton’s method was chosen for its quadratic convergence rate. Plain Newton diverges for realistic parameters because the reaction terms are strongly nonlinear; this was fixed with backtracking line search (halving the step until the residual decreases) and non-negativity projection. A further issue found later: backtracking checks only whether the residual decreased, not whether the resulting state is physically plausible. A physical plausibility bound (no concentration exceeds five times the largest boundary/feed value) was added alongside the residual test.

5.4 A second solver explored then abandoned: Picard iteration

Picard (fixed-point) iteration was implemented as an independent cross-check, freezing the reaction term and solving the resulting linear problem repeatedly. It was found to genuinely not converge for the realistic parameter set, diagnosed via eigenvalue analysis: the fixed-point iteration matrix’s spectral radius exceeds one for these parameters – a genuine mathematical property, not a coding defect. Picard remains in the codebase for milder-regime testing but is not a production solver.

5.5 A third solver: pseudo-transient continuation

A third approach marches a fictitious pseudo-time backward-Euler scheme toward the steady state, with Δt starting small (heavily damped, robust) and growing geometrically toward a plain Newton step. This regularises an otherwise-singular Jacobian, making it the natural fallback when Newton’s Jacobian genuinely becomes singular at a physically degenerate state.

This solver agrees with Newton to about one part in 10^{11} . This tight agreement is useful but must be interpreted carefully: both solvers share the same underlying reaction-assembly code, so agreement between them does not independently verify that code is correct – it only shows the two algorithms converge to the same fixed point of whatever system is actually coded. Two of the project’s most consequential bugs persisted with both solvers in tight agreement, because both built on the same flawed shared code. Agreement with a structurally independent cross-check is more decisive.

5.6 Time-stepping the bacterial equations: fully implicit backward Euler

Bacterial densities are advanced with backward Euler, every term evaluated at the new time level. An earlier semi-implicit version, evaluating total density ρ at the old time level, was found unstable at moderate step sizes: the density-dependent death term is a stabilising feedback, and treating a stabilising feedback with outdated information delays it by one step, enough to destabilise the scheme at large step sizes. The fully implicit formulation restored stability far beyond where the semi-implicit version had failed.

5.7 Advection discretisation: first-order upwinding

The crowding-movement term is advective. Upwinding was chosen over centered differencing to avoid spurious oscillations, at the cost of first-order rather than second-order accuracy – a deliberate trade-off, since oscillation-free behaviour matters more than formal order for this term. The source paper’s own numerical scheme uses the same choice.

6 The Full Chronology: Problems Found, Diagnosed, and Fixed

This section documents, in the order they occurred, every significant issue encountered – the genuine sequence, including false starts.

6.1 The density-dependent death term

The first version implemented death as a constant rate rather than density-dependent. This was found by directly reading the rendered PDF equations from the source paper – an earlier automated text-extraction pass had silently dropped the density subscript. A constant death rate places no ceiling on population growth in substrate-rich regions; density-dependent death gives the population a genuine carrying capacity. A regression test now checks that the per-step growth ratio strictly decreases under ample substrate, distinguishing density-dependent death from the constant-ratio exponential growth a constant rate would produce.

6.2 The reaction term incorrectly zeroed at the domain centre

The elliptic solver zeroed the reaction term at the grid point representing the domain centre, assuming it needed the same treatment as the outer boundary. It does not – the centre is

an ordinary interior point that merely requires a symmetric formula due to the coordinate singularity. This degraded a second-order-accurate scheme to roughly first order, visible only when reactions were active. Found via systematic grid-refinement studies; fixed by restoring the reaction term at the centre.

6.3 The approximate cell-volume near the domain centre

Even after the above fix, convergence studies showed only 1.7-1.9th order, not 2.0. Root cause: cells adjacent to the centre used the shortcut $V_i \approx r_i^p h$ for their volume, carrying 7.7% relative error at the first interior point. Fixed by computing exact cell volumes via direct integration everywhere. Mass conservation then held to machine precision and the spatial error profile flattened.

6.4 Confirming second-order accuracy despite a floating-point illusion

Repeating the convergence study in ordinary double precision still showed apparent degradation at fine resolutions, dropping to order 1.4 between the finest grids. Tightening Newton’s tolerance from 10^{-9} to 10^{-14} produced identical errors to six significant figures, ruling out under-convergence. Investigation revealed catastrophic cancellation: the benchmark’s residuals summed two terms of magnitude $\sim 10^7$ nearly cancelling to ~ 1 , leaving an absolute uncertainty of $\sim 10^{-9}$ in double precision – exactly matching the observed degradation.

To confirm rather than merely argue, the identical discretisation was re-solved with fifty-decimal-digit arbitrary-precision arithmetic via a hand-written tridiagonal solver in `mpmath` (bypassing SciPy’s locked-in double-precision LU). With floating-point roundoff eliminated, convergence order came out clean and monotonic: 1.999-2.000 across all resolutions, averaging 1.9997. This proved the discretisation is genuinely second-order correct; the double-precision degradation was entirely an artefact of the benchmark’s extreme coefficient magnitude.

6.5 Two errors in reaction-coefficient assembly

Two related errors were found in how Λ_{ij} and Λ_i^{prod} were assembled. First, Λ_{ij} already fully incorporates a species’ growth rate; an earlier version multiplied by $\hat{r}_i$ a second time when assembling the uptake term, effectively squaring the growth-rate dependence. Second, the production term for a downstream substrate (e.g. nitrite produced by AOB) reused the source substrate’s coefficient rather than a separate coefficient built with the produced substrate’s own diffusivity.

As noted above, both errors were invisible on the toy and rebecca presets and exposed only by eloi's realistic heterogeneity. The measured impact on eloi: nitrite centre concentration changed by over 100-fold once corrected, nitrate by roughly 40%. The by-product substrates, having no external feed to buffer the error, were most affected.

6.6 A boundary-condition bug affecting only nonzero flux

While exploring Neumann boundary conditions, a bug was found in three separate files: each correctly built the matrix row for the requested flux, but a separate piece of code computing the target residual had zero hardcoded into it regardless of the requested flux. Any nonzero-flux request silently produced a zero-flux solution, with no error. This is a recognisable failure mode – two independently-written pieces of code representing the same physical equation drifting apart. Fixed in all three locations; a new test checks both agreement with the analytical solution and that a finite-difference recomputation of the boundary flux from the converged profile matches the requested value to machine precision.

At the time this fix was made, it was verified only for a single substrate carrying nonzero flux while the other three used ordinary fixed-concentration boundaries. Whether the fix held when more than one substrate needed nonzero flux simultaneously was, at that point, untested – and, as described below, turned out to still be broken.

6.7 A physically implausible solution accepted by Newton

A sealed-system test case (zero flux everywhere, chemistry active) had undamped Newton converge to ammonium concentrations roughly 100-fold and nitrate roughly 5000-fold above the starting point – mathematically valid (the residual was genuinely small) but physically nonsensical, since a sealed system has no external source. The cause: backtracking checks only whether the residual decreased, with no notion of physical plausibility. A physical upper bound (five times the largest boundary/feed concentration) was added to the acceptance criterion alongside the residual test.

6.8 A self-inflicted sign error, and genuine degeneracy handled gracefully

Fixing the above introduced a smaller issue: the upper bound was computed as five times the largest *signed* value among boundary/feed concentrations, which under Neumann conditions can legitimately be negative (representing influx). Taking the maximum of a set including negative values silently collapsed the bound to zero whenever every flux was negative, clipping

every trial concentration to zero. Fixed by using the largest magnitude rather than largest signed value.

Separately, a genuinely different, non-bug situation was identified: once a substrate is legitimately driven to exactly zero everywhere – the physically correct outcome for oxygen in a sealed, fully anoxic system – Newton’s Jacobian can become exactly singular. This is now explicitly detected and handed to the pseudo-transient solver, whose regularising term can push through degeneracies plain Newton cannot.

6.9 A diagnostic-transparency problem from the fallback

Once the automatic fallback existed, the calling code logged every solve as ”Newton” regardless of whether the fallback had silently engaged – a loss of diagnostic honesty, not a correctness bug. The Newton solver now reports explicitly which method produced the result, and this propagates to the calling code’s logging. A related tolerance mismatch was fixed simultaneously: the fallback had been using its own default tolerance rather than the caller’s requested tolerance.

6.10 A duplicated-computation bug in the bacterial solver

The domain-centre cell-volume fix was correctly applied to the substrate solver’s Laplacian code, but the bacterial solver’s advection operator and its separate mass-diagnostic routine had each independently reimplemented the same already-fixed wrong volume formula. This error also affected the outer boundary in every geometry, and did not shrink with grid refinement – a structural bug, not an ordinary discretisation error. It survived earlier testing because the advection mass-conservation regression test used a probe function that happened to be exactly zero at the outer boundary, precisely where the error was largest; a different probe function immediately revealed an error 100 times larger. Fixed by creating a single shared routine for cell volumes and face areas, used by every operator that needs them, with the existing Laplacian code refactored onto it and confirmed to produce bit-for-bit identical results.

6.11 A remaining instance of the duplication pattern, now consolidated

An adversarial review, deliberately looking for further ”independently-defined duplicates” after finding the bug above, found one: the species list and each species’ primary-substrate assignment were independently defined in two separate files. Both copies happened to agree

at the time – no incorrect result occurred – but this is exactly the pattern that twice before silently drifted into real bugs. Consolidated into a single shared definition.

6.12 The closed-form benchmark used throughout

Many fixes above were verified against a deliberately constructed reduced version of the model. The full nonlinear model has no known exact analytic solution – a well-established fact about saturating reaction-diffusion equations generally, not a gap in this project’s effort. Setting two species to zero and pushing the remaining species’ half-saturation constant very large forces the Monod term into a linear regime, collapsing the substrate equation to the classical linear reaction-diffusion problem with standard exact solutions:

$$\text{slab: } c(r) = c_\infty \frac{\cosh(\kappa r)}{\cosh \kappa}, \quad (10)$$

$$\text{cylindrical: } c(r) = c_\infty \frac{I_0(\kappa r)}{I_0(\kappa)}, \quad (11)$$

$$\text{spherical: } c(r) = c_\infty \frac{\sinh(\kappa r)}{r \sinh \kappa}. \quad (12)$$

Newton and Picard solutions were compared against these formulas at several κ values; Newton matched to better than one part in 10^4 , Picard to about one part in a thousand.

7 Validation and Cross-Checking

Total mass conservation was confirmed to machine precision for all three geometries. A second, independent structural check confirmed the discrete system respects total nitrogen conservation across the three-step oxidation chain to within 10^{-16} – a physical law never used in deriving the model’s coefficients, so a genuinely independent check. The discrete system was additionally confirmed self-adjoint and to respect the discrete maximum principle.

Spatial convergence order was confirmed by grid refinement, achieving clean 2.0th order for pure diffusion and, via the fifty-decimal-digit study described above, globally for the nonlinear-reaction case as well.

Both the substrate and bacterial solvers’ full Jacobians were checked against finite-difference approximations and agreed to the precision expected of first-order finite differences ($\sim 10^{-6}$ relative).

The closed-form benchmark, extended to Neumann boundary conditions, provided the reference solution for the fifty-digit convergence study.

Newton’s method, Picard iteration, and pseudo-transient continuation were cross-compared

across all three presets and a range of bacterial densities, agreeing to roughly 10^{-11} – several orders tighter than necessary, but useful evidence only when interpreted carefully: this agreement shows the algorithms converge to the same fixed point of shared code, not that the code is correct. The reaction-coefficient bugs described above persisted through periods when all three solvers agreed to 10^{-11} , because all three called the same flawed assembly function. Agreement with structurally independent ground truth – the closed-form benchmark, mass conservation, the nitrogen-conservation law – is what provides decisive evidence.

8 The Two-Dimensional Extension

8.1 Motivation and design decision

The one-dimensional solver assumes radial symmetry, which captures zonation well but structurally cannot capture angular sector formation, a phenomenon the source paper specifically investigates. Extending to two dimensions required a genuine design choice: reproduce the source paper’s own approach (explicit time-stepping, no quasi-steady-state reduction) or extend this project’s already-validated machinery to two dimensions. The second option was chosen to preserve the entire one-dimensional validation foundation. The cost: any comparison against the source paper’s own results is necessarily between two different numerical schemes solving systems that are not quite identical, since the paper makes no quasi-steady-state assumption. This distinction turned out to matter, as discussed below.

8.2 Discretisation and validation

The two-dimensional extension uses the same finite-volume philosophy; in Cartesian coordinates there is no radial singularity to handle specially. Exact mass conservation was confirmed to machine precision. A slab-geometry reduction confirmed agreement with the already-validated one-dimensional slab solver to solver tolerance ($\sim 10^{-14}$) rather than mere discretisation-error agreement – a check that simultaneously validates the Laplacian, boundary conditions, reaction assembly, and Jacobian at once.

A second reduction check compared the two-dimensional solver, run with radially symmetric conditions, against the one-dimensional *cylindrical* ($p = 1$) solver, not the spherical ($p = 2$) solver – a distinction derived two independent ways, symbolically and by confirming numerically that only $p = 1$ ’s error shrinks under refinement while $p = 0, 2, 3$ sit at a fixed mismatch regardless of grid fineness. With the correct comparison, agreement was confirmed to genuine second order in the domain interior.

8.3 Sector formation reproduction

Using this project’s own realistic (eloi) parameters, an initial attempt did not produce visible sector formation; species grew essentially uniformly. Rather than leave this as an unexplained negative result, the source paper’s exact Table 1 parameters for its three-species competition case were used instead. With those parameters, sector formation reproduced successfully and unambiguously: three species segregated into distinct angular wedges with near-complete mutual exclusion (dominant species density ~ 0.047 in its wedge, excluded species ~ 0.000016 in the same location – a difference of over 3000-fold), the colony expanded outward while its interior decayed from substrate depletion, consistent with the paper’s description.

An important discrepancy surfaced. Computing the scale-separation parameter ε for the paper’s own Case A parameters gives $\varepsilon \sim 10^4$ – substrate diffusion is roughly 10000 times *slower* than bacterial growth, the direct opposite of the regime this project’s entire quasi-steady-state strategy depends on. The reproduction was obtained by applying the solver outside its own regime of validity. This is recorded explicitly as an open question: whether the quasi-steady-state assumption remains reasonable in this high-Damkohler regime has not been independently verified.

8.4 Investigating the sector-formation transition

A natural follow-up: what quantitatively determines the transition between sector formation and uniform growth? An initial hypothesis – that diffusion length versus domain size controls the transition – was tested by searching for the specific bacterial diffusivity at which behaviour switches.

An initial search, at a single coarse grid resolution, located the transition at $d_i \approx 4.9 \times 10^{-4}$. The naive hypothesis predicted a transition around $d \approx 0.067$ – roughly two orders of magnitude off, in the opposite direction. This was investigated in three steps rather than reported as a bare gap.

First, whether a different length scale could rescue the naive hypothesis was checked: the controlling dimensionless number was evaluated at the measured threshold for domain size ($Da \approx 2054$), inter-seed spacing ($Da \approx 139$), seed radius ($Da \approx 20.5$), and grid spacing ($Da \approx 2.6$, as a reliability check rather than a physical candidate). That grid spacing came closest to order one was a warning sign of a possible resolution artifact, not evidence for grid spacing being the right physical scale.

Second, that warning sign was tested directly: the search was repeated at two finer, fixed resolutions. If the threshold had been a pure discretisation artifact, it should have kept shrinking proportionally to grid-spacing-squared. Instead it converged to $d_i \approx 2.7 \times 10^{-4}$

and stayed there at finer resolution still – ruling out ”pure grid artifact” while confirming the original coarse estimate had been biased roughly 1.8-fold high. Re-evaluated at this corrected threshold: domain size gives $Da \approx 3652$, seed spacing $Da \approx 246$, seed radius $Da \approx 36.5$ – none reaching a clean order-one value, though seed radius comes closest.

Third, whether bacterial diffusivity alone controls the transition was checked by fixing d_i at the threshold and instead varying the crowding-movement ratio a_i/d_i across 3, 10, 30, 100. The transition location shifted measurably, confirming the mechanism depends on at least two parameters together.

Taken together: the existence of a genuine, resolution-independent transition is established. The naive domain-size-based diffusion-length explanation is ruled out; no closed-form scaling law has yet replaced it. The seed-radius length scale came closest to an order-one number, pointing toward this being a pattern-selection question – whether the seeded angular perturbation grows or decays – rather than a simple diffusion-length balance. The linear stability analysis that would settle this precisely has not been carried out and is recorded as open work.

9 Closing the Multi-Substrate Neumann Gap

The boundary-condition bug fix described above left one entry point genuinely unreliable: requesting nonzero flux simultaneously on more than one substrate. This section describes how that gap was closed, what two further bugs were found in doing so, and what genuine limitation remains.

9.1 The architectural problem

The boundary-condition interface originally accepted a single type applied globally to all four substrates – workable when every substrate uses the same treatment (as all production results in this document do), but physically nonsensical the moment two substrates need genuinely different treatments at once (e.g. ammonium held at a fixed external concentration while nitrate, with no external supply, needs a fixed flux instead). Forcing both into the same global type meant any attempt at independent nonzero flux on more than one substrate at once was fragile.

9.2 The refactor

The boundary-condition interface across all three elliptic solvers was extended to accept an independent specification – boundary type and value – for each of the four substrates,

rather than one shared specification for all four. The previous single-global-type interface was confirmed, by rerunning the entire existing test suite unchanged, to produce identical results under the new machinery – the old behaviour is simply the special case where every substrate happens to share the same specification. The two-dimensional solver received the equivalent extension, with one honestly stated scope limitation: two-dimensional Neumann conditions still support only zero flux.

9.3 Two further bugs found while building the fix

Building a genuine test of coupled nonzero flux on two substrates simultaneously surfaced two further, previously invisible bugs. First, the solver’s default starting guess for any flux-boundary substrate was zero; since the Monod function is exactly zero at zero argument, starting a co-limiting substrate like oxygen at exactly zero silently zeroed every species’ entire growth term and reaction Jacobian at the first iteration – a self-inflicted version of the same degeneracy already identified elsewhere. Fixed by starting flux-boundary substrates at a small positive reference value. Second, the physical-plausibility upper bound had only ever been added to Newton’s own backtracking loop; the pseudo-transient fallback had a lower (non-negativity) bound but no upper bound, and was observed to diverge to concentrations around 3.7×10^{11} in one constructed case. The same upper-bound guard was added to the fallback.

9.4 Result and honest remaining limitation

With both fixes in place, a genuinely coupled two-substrate nonzero-flux configuration, at moderate reaction-rate coefficients, converges cleanly at multiple grid resolutions, recovered boundary flux matching the requested value to better than one part in 10^{15} .

The same configuration at the eloi preset’s actual reaction-rate stiffness does not converge. Rather than continue attempting fixes indefinitely, the Jacobian at the stall point was examined via singular value decomposition and found genuinely, severely near-singular – condition number of order 10^{16} , at the edge of double-precision representability. This is reported as a confirmed numerical-conditioning limitation of the realistic parameter regime itself, not a remaining architectural gap: the boundary-condition machinery works correctly and generally; the specific combination of realistic stiffness with coupled nonzero-flux boundaries remains numerically difficult.

10 A Direct Quantitative Comparison Against the Source Paper

10.1 Reproducing the paper’s travelling-wave result

The source paper reports a specific, directly reproducible result: for a parameter-symmetric case reducing to total biomass and a single substrate, a measured travelling-wave speed compared against a closed-form minimal-speed formula. This scenario was implemented independently using the paper’s own Table 1 parameters. The measured propagation speed (tracking the biomass-density peak over time, as the paper does) came within approximately 0.85% of the paper’s reported value. The minimal wave speed formula, re-derived independently via symbolic computation, agreed with the paper’s reported formula value to within better than 2%.

This agreement is meaningful because this project’s numerical approach (quasi-steady-state elliptic reduction) is genuinely different from the source paper’s (direct explicit Runge-Kutta integration). Two structurally unrelated methods agreeing this closely on an independently measured quantity is non-circular evidence that both correctly solve the underlying model.

10.2 Re-verification with completely independent implementation

As a further adversarial check, the comparison was repeated with a completely separate implementation, reusing no code from the main solver, with the minimal-speed formula re-derived symbolically from first principles using a computer algebra system rather than re-typed from the paper. This independent implementation agreed even more closely with the paper’s value – within roughly 0.25% rather than 0.85% – reinforcing that the agreement is a genuine property of the models and methods, not an implementation artifact.

The same adversarial approach was applied to one further internal claim: the exact mass-conservation identity underlying the two-dimensional finite-volume Laplacian was re-derived analytically from scratch and re-checked numerically with an independently hand-written stencil, unconnected to the project’s own grid code, across several grid sizes and random test fields. All checks landed at machine-precision zero.

11 Present Status and Open Items

11.1 What is firmly established

The one-dimensional solver’s mathematical structure has been verified independently, not assumed correct because the code runs without crashing. Full Jacobians have been checked against finite-difference approximations to expected precision; the discrete system conserves total mass to machine precision, and total nitrogen exactly across the oxidation chain to within 10^{-16} , using a law never used in deriving the model, ensuring a genuinely independent check; and spatial second-order accuracy has been proven with floating-point roundoff entirely excluded via fifty-decimal-digit arithmetic.

Nine genuine, consequential bugs were found and fixed: a foundational death-term error; incorrect zeroing of the reaction term at the domain centre; insufficient cell-volume accuracy near the centre; hardcoded incorrect Neumann targets in three locations; acceptance of a physically absurd solution by Newton; duplicated, inconsistent cell-volume calculations in the bacterial solver; two separate reaction-coefficient assembly errors; an unsafe flux-boundary starting guess; and a missing plausibility bound in the fallback solver.

The originally open multi-substrate Neumann limitation has been closed for moderate reaction stiffness, with the remaining limitation at realistic stiffness now understood precisely as a confirmed near-singular Jacobian rather than an architectural gap.

The two-dimensional extension has been validated independently and has reproduced the source paper’s specific qualitative phenomenon using the paper’s own parameters. A direct, independently reproduced quantitative comparison against one of the paper’s own numerical results agrees to within roughly a quarter of one percent, using an implementation sharing no code with the main solver.

The full test suite passes in its entirety, and everything described here, together with this document itself, is committed to version control.

11.2 What remains open

The complete mechanism controlling the sector-formation transition depends on at least two parameters, and a simple diffusion-length-versus-domain-size explanation has been definitively ruled out by direct grid-refinement testing – but no closed-form scaling law has replaced it. Linear stability analysis of the seeded angular perturbation is the identified next step, not yet carried out.

The sector-formation reproduction was obtained in a parameter regime where the quasi-steady-state assumption anchoring this project’s entire numerical strategy is not obviously

appropriate ($\varepsilon \sim 10^4$ rather than $\ll 1$). Whether that reduction remains reasonable in this regime has not been separately investigated.

Picard iteration does not converge for the realistic parameter set without a workaround, confirmed to be a genuine mathematical property (spectral radius exceeding one) rather than a coding defect; it is not relied on as a production solver.

The multi-substrate Neumann interface converges reliably at moderate stiffness but not at realistic stiffness, confirmed to be genuine numerical conditioning rather than a missing capability. This does not affect any production results, all of which use fixed-concentration boundaries throughout.

The decision to divide rebecca’s originally-supplied yield values by one hundred remains a genuine modelling choice taken on faith, now shown by sensitivity analysis to be load-bearing in a precise sense: solver convergence, the direction of mass growth, and species dominance are unaffected by the choice, but whether the anoxic interior zone forms at all depends materially on it. Resolving this properly would require returning to the original source to clarify intended units.

No software licence currently exists in the repository; citation metadata deliberately omits a licence field rather than assuming one. Robin boundary conditions, identified by both the source paper and this project as the eventual appropriate target, have not yet been implemented in either solver.

12 File-by-File Guide to the Codebase

`params.py` holds the three dimensional parameter presets, together with the documented reasoning behind every cleaning decision – including the rebecca divide-by-100 yield rescale and the constants used to derive the packed-biomass reference density.

`nondim.py` performs the non-dimensionalisation: reference scales, dimensionless groups, and $\Lambda_{ij}/\Lambda_i^{\text{prod}}$, the coefficients at the centre of one of the project’s consequential bugs.

`elliptic.py` implements the one-dimensional quasi-steady substrate solver: the finite-volume grid and Laplacian, handling the radial singularity; the shared cell-volume and face-area routines; boundary-condition machinery for both Dirichlet and Neumann conditions, including the physical-plausibility guard and the per-substrate boundary specification; Monod reaction-term assembly; and both solution methods. The hardcoded-zero Neumann-target bug and the flux-boundary starting-guess bug were both fixed here.

`relaxation.py` implements the pseudo-transient continuation solver, used both as an independent cross-check and as the automatic fallback for genuine degeneracies. The missing upper-bound guard was fixed here.

`parabolic.py` implements the one-dimensional bacterial solver: the fully implicit backward-Euler scheme, the advection operator (subject to the duplication bug), the density-dependent death and growth terms, and the cross-diffusion Jacobian. Also contains the shared cell-volume routines after consolidation.

`slowfast.py` implements the overall simulation loop, alternating between `elliptic.py` (or its `relaxation.py` fallback) and `parabolic.py`, and is responsible for the diagnostic method-labelling that reports which solver actually produced each result. This file always uses fixed-concentration boundaries and was deliberately not extended with the per-substrate interface, since doing so would add surface area without adding capability actually used in the production pipeline.

`grid2d.py`, `elliptic2d.py`, `parabolic2d.py`, and `slowfast2d.py` are the two-dimensional counterparts, sharing the same finite-volume philosophy, cell-volume consolidation, and (for the elliptic solver) the per-substrate boundary-condition interface, with two-dimensional Neumann support limited to zero flux.

The `tests/` directory contains one file per major solver component, plus dedicated files for the reaction-coefficient assembly checks and the closed-form benchmark comparisons. The advection mass-conservation regression test's original probe function – exactly zero at the boundary, masking the duplication bug – was corrected to a nonzero-boundary probe that exposes it directly. The two-dimensional test file includes the mass-conservation and reduction checks, plus regression coverage for the mixed-boundary-condition capability.

The repository additionally contains an audit checklist cross-referencing every claim in this document against the specific test or script that verifies it; a set of standalone, runnable scripts reproducing each quantitative claim in the later sections of this document (the sector-formation threshold investigation, the travelling-wave benchmark, and the independent re-derivation checks) together with captured output; a README describing how to reproduce every reported result from the repository alone; and a citation file identifying the specific repository version used to produce the results in this document, which deliberately omits a licence field pending that decision being made explicitly.